Ibrahim Boubakri

Full-Stack Developer | DevOps | Software Development | AI Integration

+1 (819) 992-2001 ibrahimboubakri1@gmail.com ibrahimboubakri boubakriibrahim ibrahimboubakri.me

Profile

Full-stack developer focused on building reliable, high-performance, and maintainable software, with hands-on experience across backend, frontend, APIs, automation, DevOps, and AI integration. Builds modular services, containerized workflows, CI/CD pipelines, and AI/LLM-enabled application components from technical design through deployment and observability. Strong in Python, FastAPI, React, Node.js, Docker, Kubernetes, SQL, and Linux, with a consistent focus on performance, reliability, developer efficiency, scalability, and continuous improvement.

Education

Université du Québec à Trois-Rivières – UQTR **Sep. 2024 – Aug. 2026**
Master's in Electrical Engineering, Computer Engineering Concentration *Trois-Rivières, QC, Canada*

National School of Computer Science – ENSI **Sep. 2020 – Aug. 2023**
Engineering Degree in Computer Science *University of Manouba, Tunisia*

Professional Experience

Full-Stack Developer – Architecture, DevOps and AI Integration **Sep. 2024 – Present**
Modular AI Platform *Quebec City, QC, Canada / Remote*

— Contributed to the design and development of a modular software platform for workflow orchestration, data management, and AI service integration.
— Designed backend components and REST APIs for AI pipelines, structured files, metadata, execution logs, and validation mechanisms.
— Supported architectural decisions that decoupled services, improved code maintainability, and enabled modules to evolve independently.
— Integrated AI models and services into reproducible application workflows with input, output, error, and execution-status management.
— Implemented and improved DevOps practices using containerization, continuous integration, automated testing, and deployment workflows.
— Developed storage, progress-tracking, and asynchronous-processing capabilities to improve platform reliability and observability.
— Contributed to performance optimization, technical incident resolution, and documentation of software components and interfaces.
— Accelerated development and incident resolution with agentic coding assistants (Claude Code, OpenAI Codex, GitHub Copilot, and Cursor) for code analysis, refactoring, test generation, and documentation, with human review and CI/CD validation.
— *Technologies:* Python, FastAPI, REST APIs, Docker, Git, GitLab CI/CD, SQL, JSON, CSV, PyTorch, OpenCV, Linux

Researcher – Computer Vision and Autonomous Robotics **2024 – 2026**
UQTR | Industry Collaboration with Noovelia *Trois-Rivières, QC, Canada*

— Designed and developed an end-to-end computer-vision perception pipeline for industrial pallet detection and geometric keypoint estimation, with emphasis on robust and efficient real-time inference.
— Built an automated Blender-based synthetic-data workflow with domain randomization and automatic annotations to accelerate experimentation and improve generalization to real deployment environments.
— Trained, fine-tuned, and benchmarked YOLO-Pose models using PyTorch, Ultralytics, CUDA, and NVIDIA GPU infrastructure; structured reproducible evaluations for robustness, generalization, and performance/latency trade-offs.
— Developed an integrated AI and robotics platform connecting synthetic-data generation, model training, inference, PyBullet simulation, FastAPI/WebSocket services, and React to unify workflows and reduce manual operations.
— Integrated the perception module with ROS 2 and an existing navigation pipeline, deployed the solution on NVIDIA Jetson, validated real-time operation, and prepared a reusable integration pipeline for the industrial partner.
— *Technologies:* Python, PyTorch, Ultralytics, CUDA, Blender, PyBullet, ROS 2, FastAPI, WebSocket, React, NVIDIA Jetson

Developer – DevOps and Automation **Nov. 2023 – Aug. 2024**
PRIMATEC Engineering *Sfax, Tunisia*

— Developed software tools and automation scripts that reduced manual work across validation, testing, and quality-control processes.
— Integrated automated tests into CI/CD pipelines to improve the reliability, reproducibility, and speed of software deliveries.

— Designed modular Python components with structured error handling, logging, traceability, and clear separation of responsibilities.
— Maintained and improved continuous-integration environments using Docker, Jenkins, GitLab, and automation scripts.
— Implemented technical-log collection and processing mechanisms to support troubleshooting and execution monitoring.
— Collaborated with development and validation teams to standardize technical procedures and strengthen delivery workflows.

— Produced installation, usage, and maintenance documentation for internally developed tools.
— *Technologies:* Python, Docker, Jenkins, GitLab CI/CD, Jira, Batch, PowerShell, automated testing, Linux

Full-Stack Developer Intern – Web Applications and Data **Mar. 2023 – Aug. 2023**
Hydrogen Research Institute *Trois-Rivières, QC, Canada*

— Designed and deployed a microservices-based web platform for collecting, processing, and visualizing real-time technical data.
— Developed frontend and backend services and implemented interfaces for communication between platform components.
— Built automated ETL pipelines to collect, transform, and store data streams transmitted through MQTT.
— Containerized services to improve portability, isolation, and environment reproducibility.
— Supported the adoption of CI/CD practices for automated build, testing, and deployment stages.
— Contributed to data modeling, database management, and technical architecture documentation.
— *Technologies:* React, Node.js, MongoDB, Python, MQTT, Docker, Kubernetes, Jenkins

Selected Projects

Modular AI Integration and Automation Platform **Apr. 2026 – Present**

— Designed a modular architecture combining backend services, REST APIs, automation, and AI components.
— Developed workflows that orchestrate multiple processing, validation, and result-generation stages.
— Implemented execution tracking, metadata management, parameter versioning, and centralized output management.
— Integrated generative AI models into reproducible software processes with response validation and error handling.
— Containerized services and automated testing and deployment stages to simplify platform operation.
— *Technologies:* Python, FastAPI, Docker, REST APIs, SQL, GitLab CI/CD, generative AI, LLM integration, RAG, MCP, tool/function calling

HackZone Tunisia X | *Infrastructure and Cybersecurity* **Apr. 2022**

— Helped organize a 24-hour capture-the-flag event involving more than 100 teams and 300 participants.
— Contributed to the design and deployment of the technical infrastructure used to host competition challenges and services.
— Prepared technical challenges covering web vulnerabilities, Linux systems, and reverse engineering.
— Provided technical support, service monitoring, and incident resolution throughout the event.
— *Technologies:* Azure, Python, Kali Linux, Git, Docker, web cybersecurity

Technical Skills

Full-Stack and Backend Development: Python, JavaScript, React, Node.js, FastAPI, Flask, REST APIs, WebSocket, asynchronous services, authentication, systems integration, reusable component design

Software Architecture: modular architectures, microservices, API design, separation of concerns, service integration, scalability, maintainability, performance optimization

DevOps and Software Delivery: Git, GitLab CI/CD, Jenkins, Docker, Kubernetes, Linux, containerization, automated testing, deployment pipelines, observability, logging, incident diagnostics

AI and LLM Integration: generative AI, local and external LLM integration, RAG, MCP, tool/function calling, agentic workflows, response validation, PyTorch, OpenCV, AI-assisted automation

Computer Vision and Robotics: Ultralytics, YOLO-Pose, CUDA, ROS 2, NVIDIA Jetson, Blender, PyBullet, robotic perception, real-time inference, synthetic data workflows

Data and Pipelines: SQL, MongoDB, MySQL, ETL, MQTT, JSON, CSV, data validation, metadata, structured-file processing, log analysis

Programming Languages: Python, SQL, JavaScript, Java, PHP, C++, C#, PowerShell, Bash

AI-Assisted Development Tools: Claude Code, OpenAI Codex, GitHub Copilot, Cursor, code analysis, refactoring, test generation, documentation, human-reviewed validation

Languages: French fluent, English professional working proficiency, Arabic native

Certifications and Activities

— **DevOps Bootcamp** – TechWorld with Nana
— **AWS Cloud Technical Essentials** – Coursera
— **IBM DevOps and Software Engineering Professional Certificate** – Coursera, in progress
— Organizing team member for the ENSI Annual Forum, TuniHack, RoboCup, and HackZone Tunisia.